

High-precision photometry for LSST and ZTF : atmosphere gray extinction characterization with a long-wave infrared instrument within the StarDICE metrology experiment

K. Sommer, B. Plez, J. Cohen-Tanugi and the StarDICE collaboration

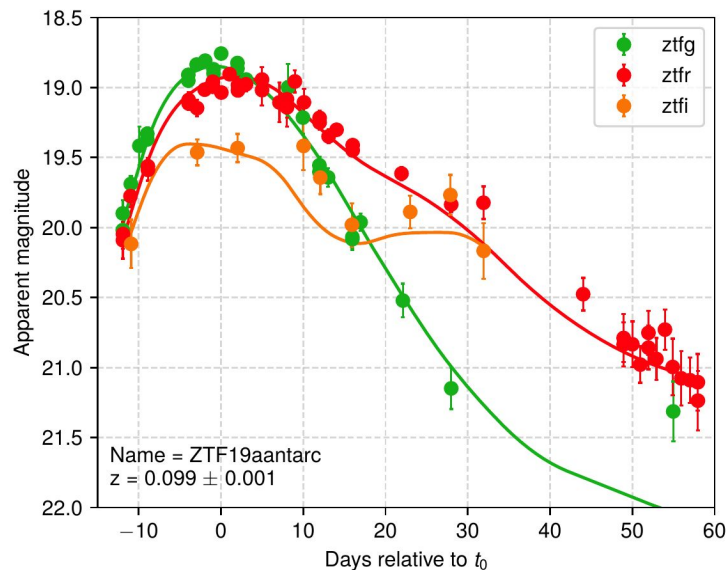
Presented by Kélian Sommer - PhD student @ LUPM CNRS/IN2P3, France

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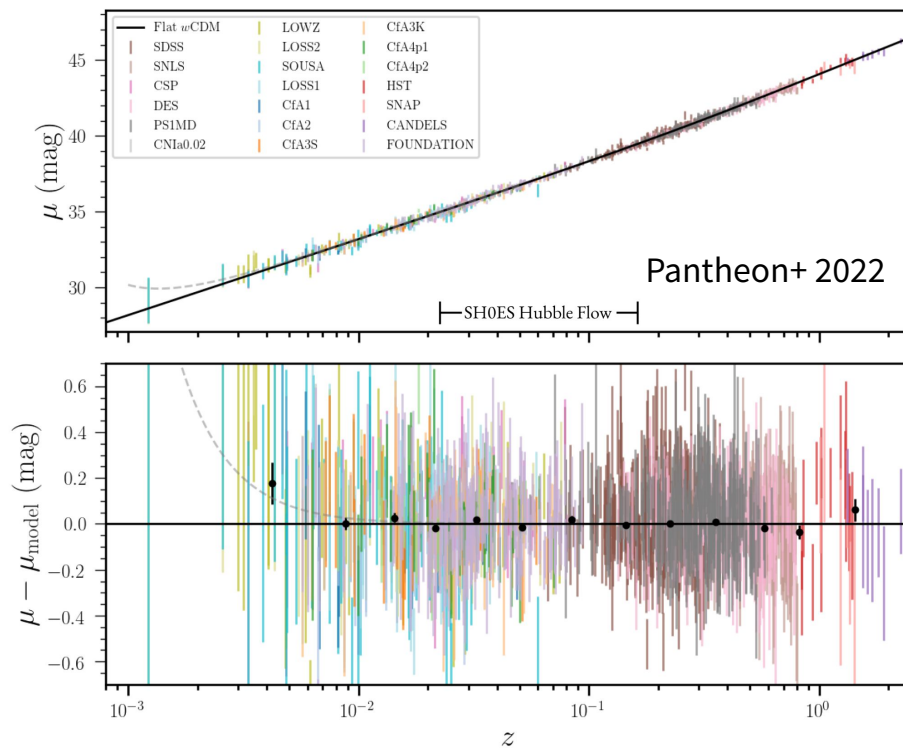


■ Cosmological context (1/2)

- Type 1a supernovae probe the recent Universe : $0 < z < 1.5$
- Rely on photometry $\rightarrow$ lightcurves
- + spectroscopy $\rightarrow$ *redshift*

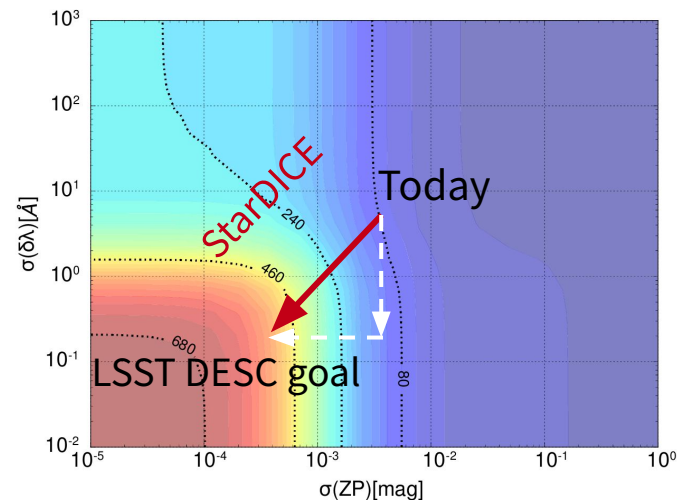
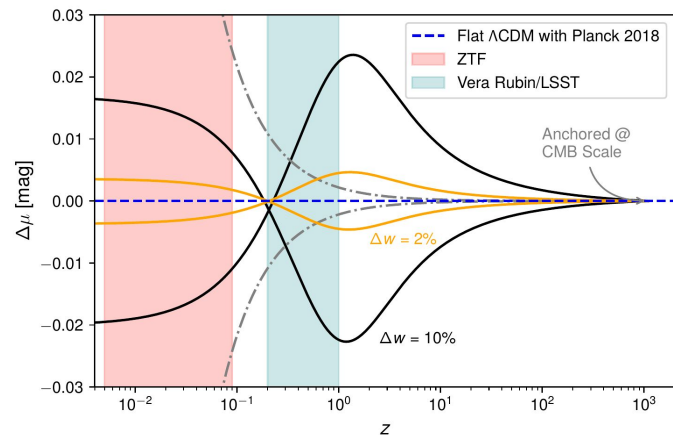


$\rightarrow$ Fit cosmological model and plot the Hubble diagram



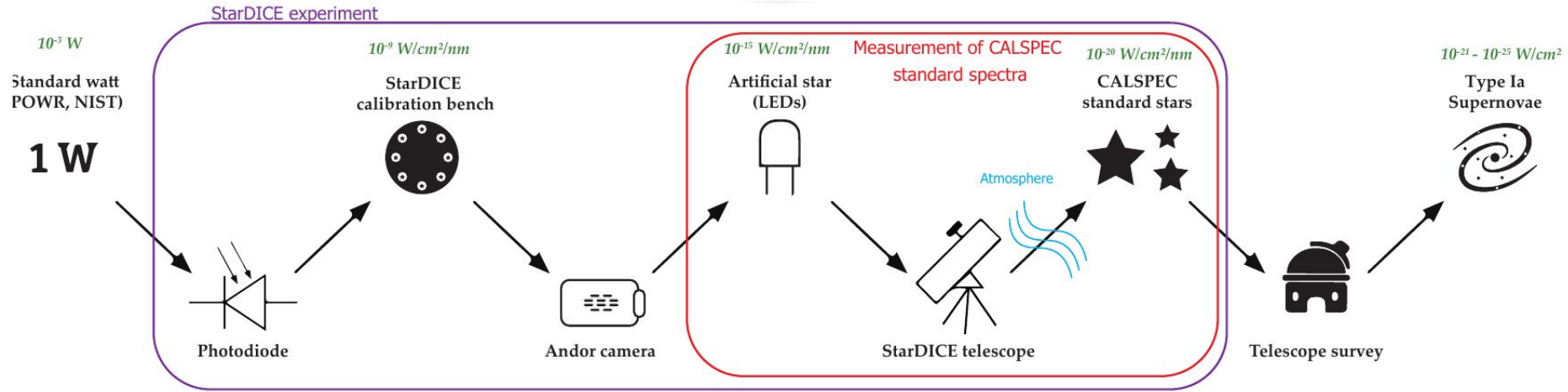
● Cosmological context (2/2)

- SNe Ia cosmology limited by:
 - Statistics
 - **Systematics** (models, **calibration**, host effects...)
- Advent of Stage IV surveys such as LSST & ZTF resolve the statistical issue:
 - ZTF (2017-2023) ~ 3800 SNe Ia
 - LSST (2025+) ~ 40,000 SNe Ia
- Improved constraints for dark energy measurements with SNe Ia require:
 - $\sigma(z_p) \leq 1 \text{ mmag}$
 - $\sigma(\lambda_{\text{eff}}) \leq 0.1 \text{ nm}$
- Best available calibration to this date → **HST/STIS + pure-hydrogen WD models**
~ 10 mmag absolute



StarDICE metrology experiment (1/2)

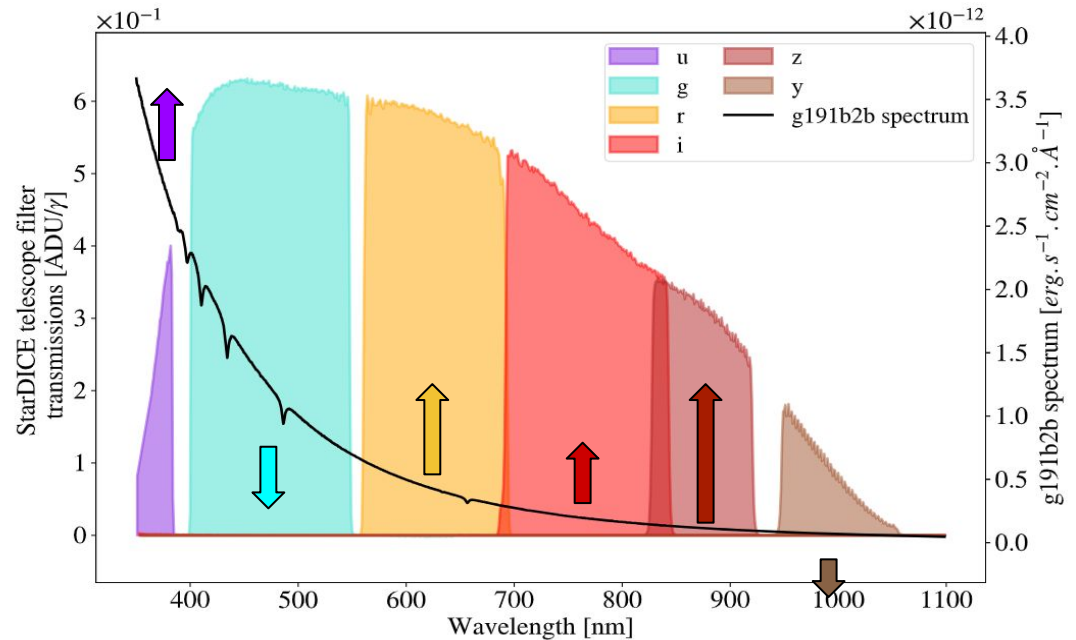
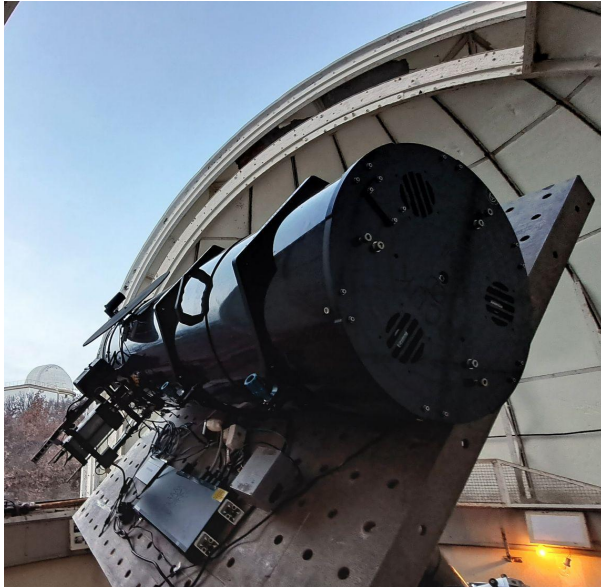
- **Goal:** to provide a measurement of the *ugrizy* colours of 3 WD pure hydrogen standard spectrophotometric calibration stars at 1 mmag or better.
- **Method:** development of complete instrumentation transferring the absolute power calibration established by NIST to a standard observing system using artificial stars.



StarDICE metrology experiment (2/2)

- Prior spectra given by CALSPEC models
- Prior knowledge of filter transmissions (CBP + artificial stars)
- Prior atmosphere simulation (using RTE)

⇒ **Combination of observations and forward modeling to adjust the spectrum for each filter**



“Gray” extinction as a limiting systematic (1/2)

CBP instrument measures
instrumental transmission

$$F_X = A_{tel} \times t_{exp} \times \int_{\lambda} S_*(\lambda) \times T_X(\lambda) \times T_{atm}(\Theta) \times \frac{\lambda d\lambda}{hc}$$

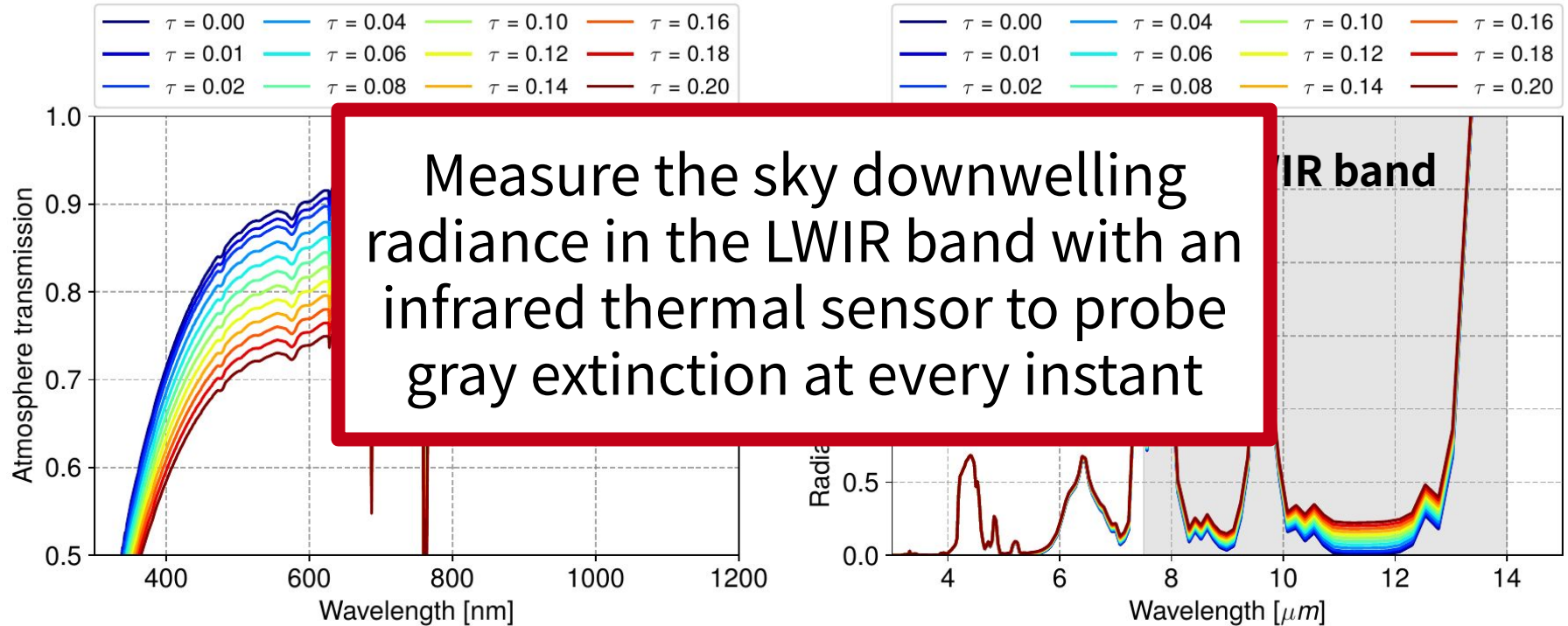
Forward model : local measurements +
radiative transfer simulations

$$T_{atm}(\Theta) = T_{gray}(t, alt, az) \times \frac{T_{mol}(P_{atm}, t, alt, az, \lambda, O_3)}{\times T_{PWV}(PWV, t, alt, az, \lambda) \times \exp(-X \times \tau_{aer}(t, \lambda))}$$

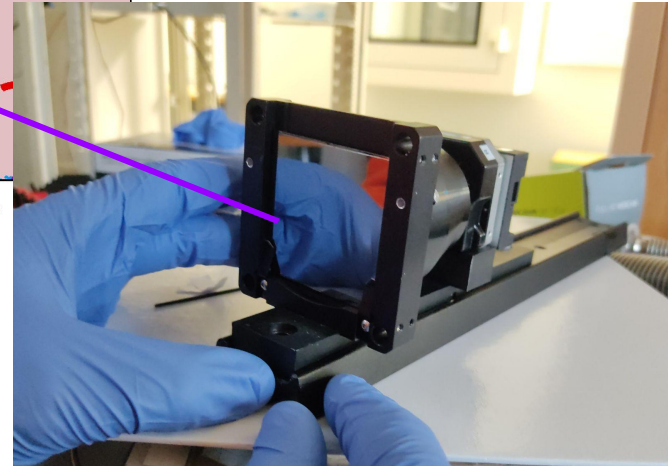
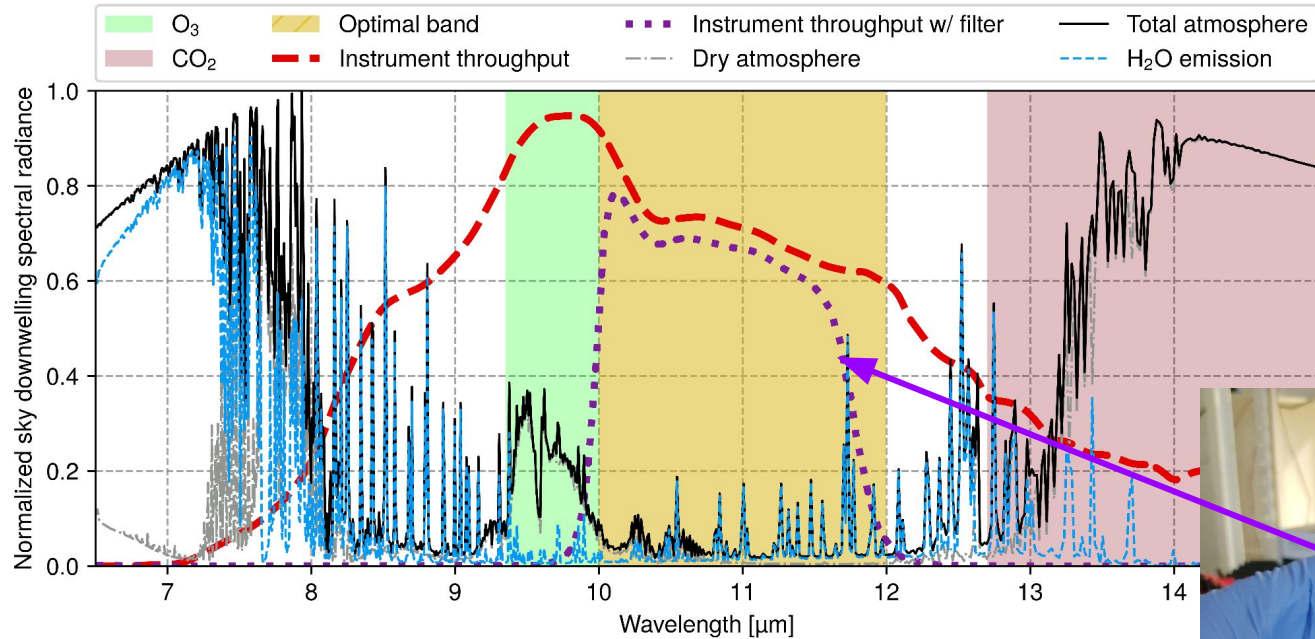
- Burke et al. 2010 → fit of T_{gray} is insufficient as calibration is limited by a rapidly changing conditions with moving and evolving cloud cover
- Hazenberg 2019 → “**last dominant contribution is gray extinction on telescope line of sight**” for StarDICE pathfinder

“Gray” extinction as a limiting systematic (2/2)

→ Measuring the sky radiance in LWIR band appears as a potential solution

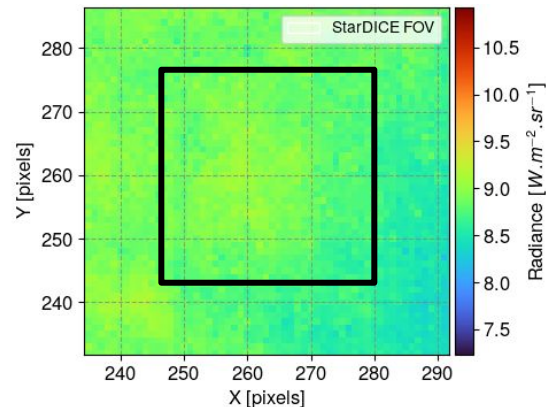
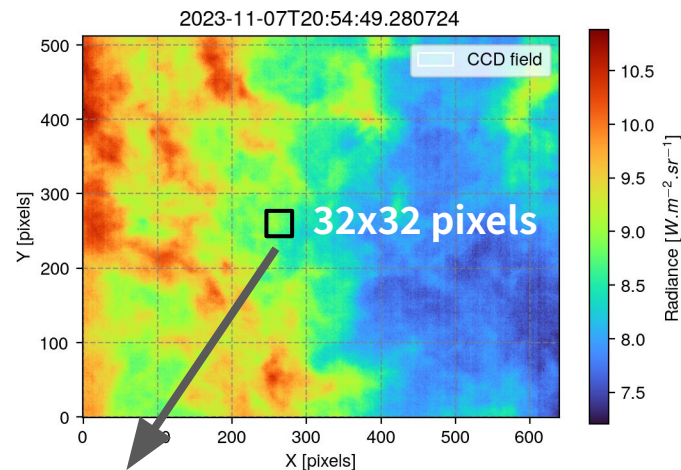
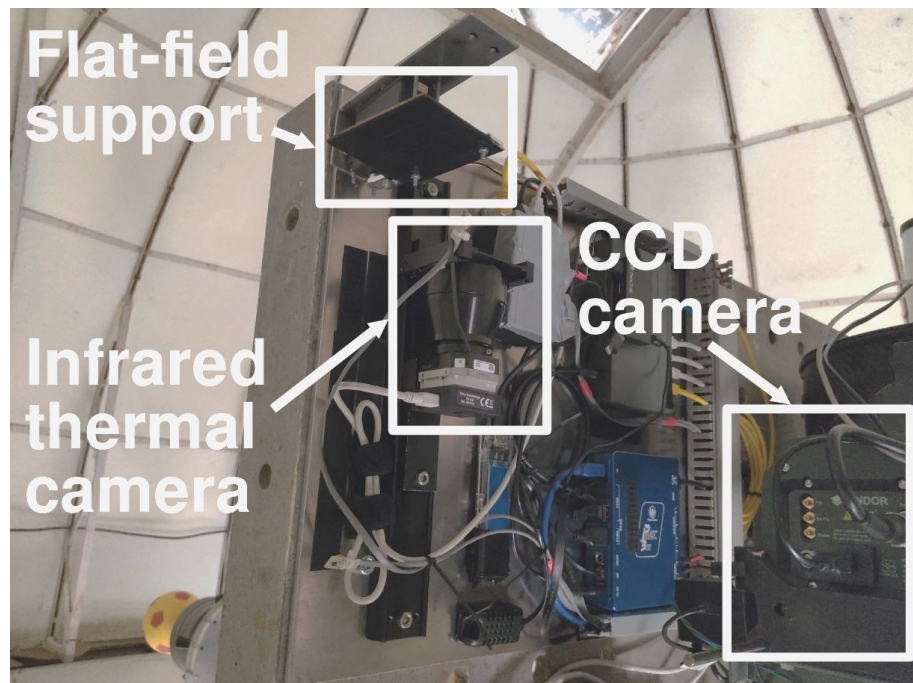


● Infrared thermal camera

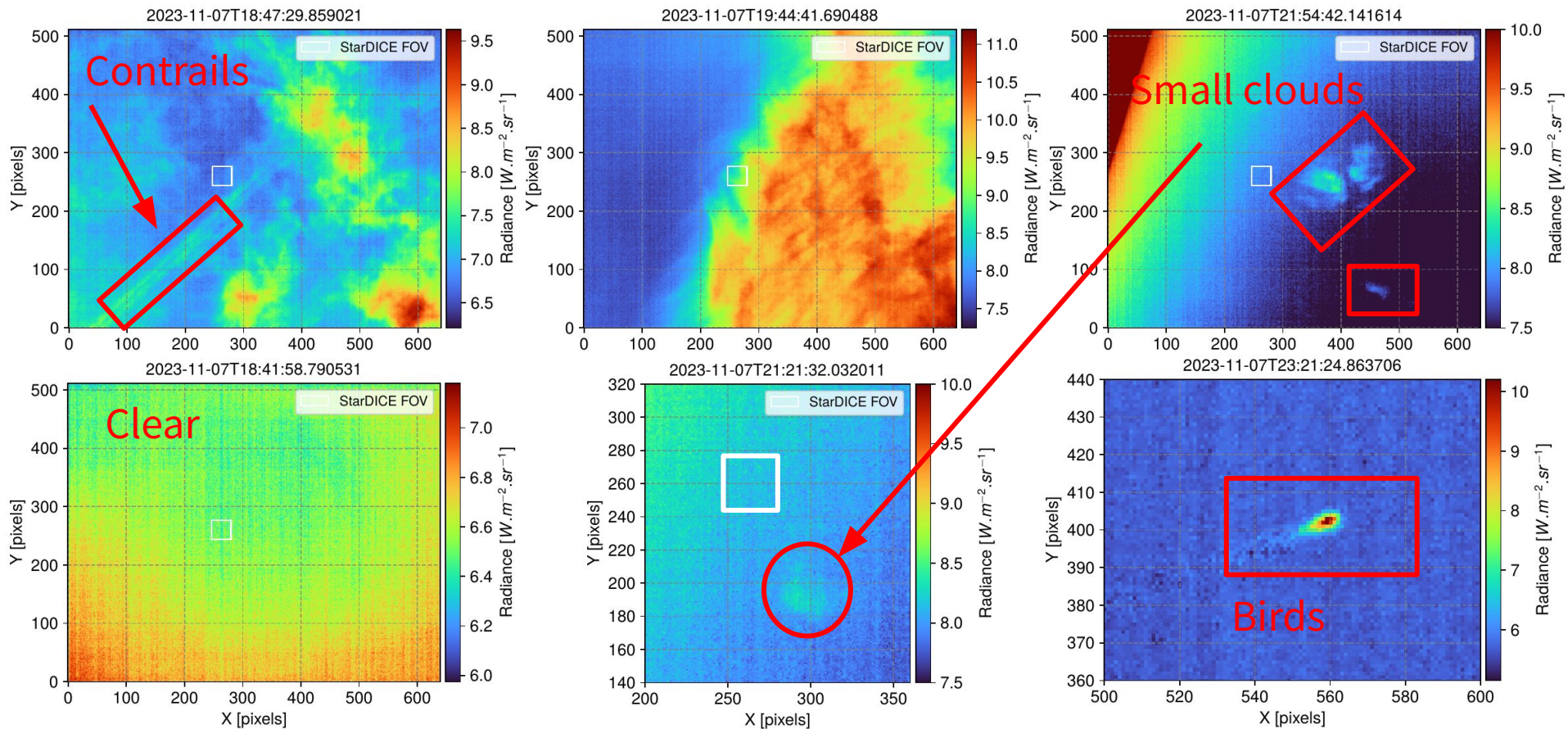


- 640 x 512 pixels uncooled microbolometers focal plane array
- Large FOV = 10.4 x 8.3 deg
- 8.33 Hz imaging rate $\Rightarrow$ ~ 80 GB/night
- Radiometrically calibrated radiance images in $\text{W/m}^2/\text{sr}$

Installation at Observatoire de Haute-Provence



Quick look : a lot of things happen in 4h of imaging...



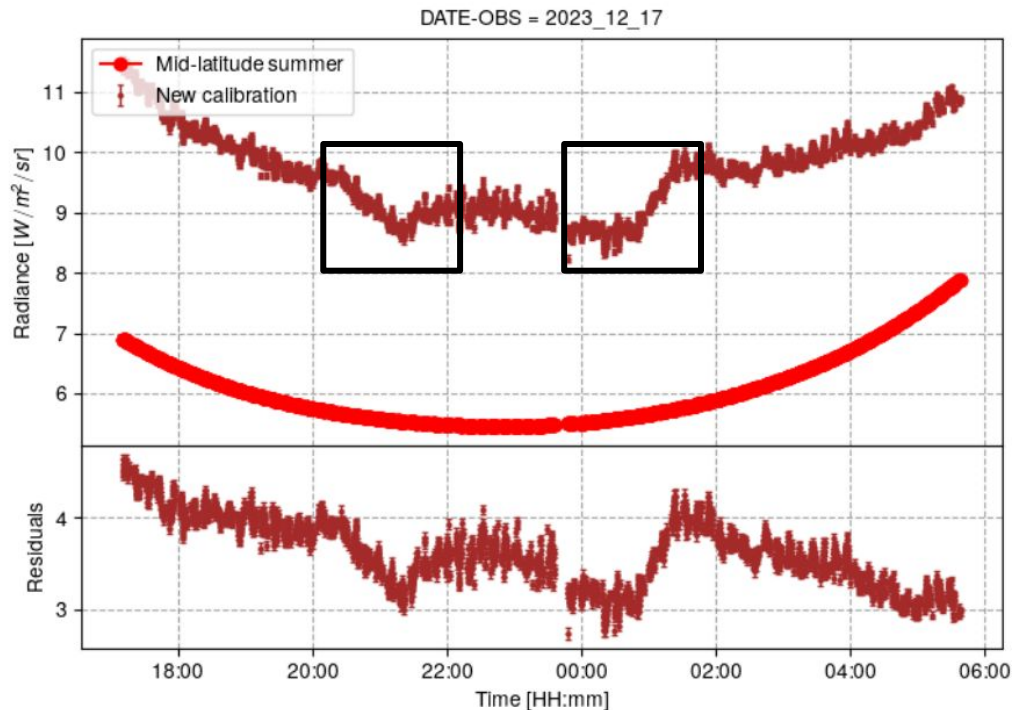
Major issue

- Dome is thermal emitter itself at ~ ambient temperature
- Variations of dome position relative (known to ~ 1 deg max.) to mount positions

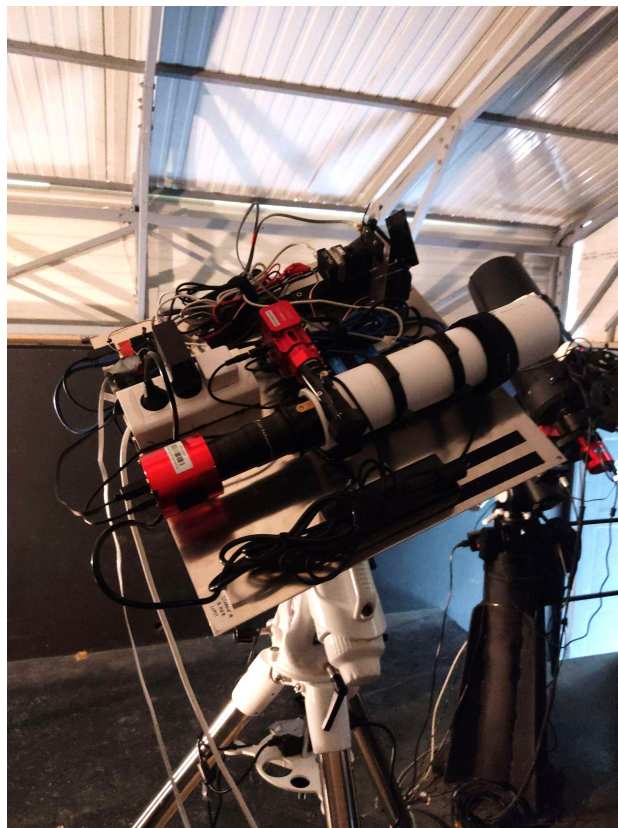
⇒ variations in the background level and structure/gradient in images

⇒ challenge to remove the “True” sky background emission to retrieve cloud(s) structure(s) and optical depth(s)

- **Mitigation plan** : move the IR instrument inside a “roll-off roof” type building

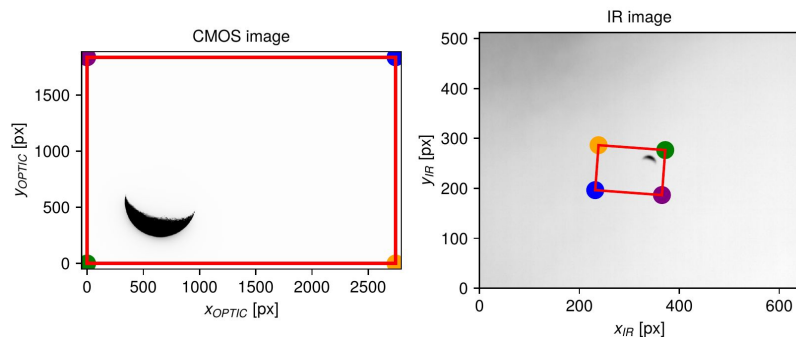


● New setup at Satino building



Good news and future

- Remote operations since May
- ~ 6 nights of data:
 - +1700 *r*-band images
 - +90,000 IR images
- Matched FOVs with success
- No gradient observed
- Analysis in progress...



Preliminary analysis 2024-05-30 on CALSPEC HD180609

